

TOUCH TO ROLL: A Decoder-Based Dice

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Electronic dice, or e-dice, are widely used today, but creating a design replicating the anticipation and randomness of a physical dice roll comes with its challenges. This device achieves that effect by

generating a rolling motion that lasts approximately four seconds, though the duration can be adjusted by modifying the rolling steps.

A 3-to-8 decoder (IC 74LS138) displays a valid dice face, with combinational AOI logic (AND, OR, and INVERT gates) ensuring only valid outputs. Seven individual LEDs, arranged to represent numbers 1 through 6, replicate the appearance of a traditional dice. The ESP32 microcontroller brings the design to life, handling random number selection and creating a realistic rolling effect, making this an excellent case study in digital circuit design.

Fig. 1 shows the prototype with the rolling clock, while Fig. 2 presents a block diagram of the system. Table 1 lists the com-

ponents required to build this device.

The main components of this device are the ESP32 board and the ICs 74LS138, 74LS260, 74LS04, and CD4075. The truth table for the decoder IC 74LS138 is provided in Table 2.

The ESP32 board is the core of the device. It is used for the following tasks:

Detecting the closure of the path that activates the dice roll.

Introducing a rolling step of a few seconds (four seconds in this case).

Generating a 3-bit random number.

This 3-bit number is sent to a 3-to-8 decoder circuit modified to meet the dice roll requirements. Due to logical modifications, its truth table is adapted for dice use. The 74LS138 is a negative logic activated IC, meaning its output lines switch to a low state instead of a high state when activated.

For proper operation:

Enable pins G2A' and G2B' must be connected to logic low.

Enable pin G1 must be connected

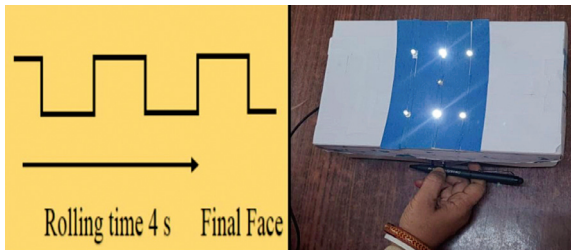


Fig. 1: Authors' prototype with rolling clock and e-dice rolling effect

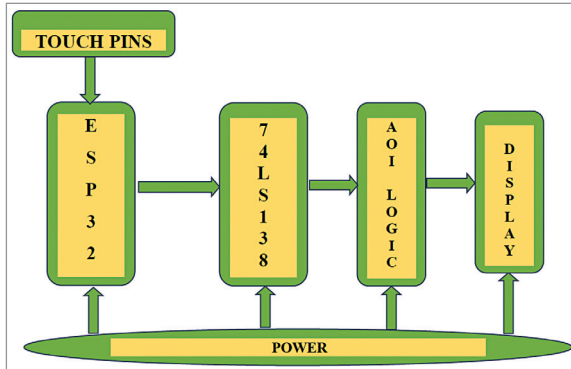


Fig. 2: Block diagram

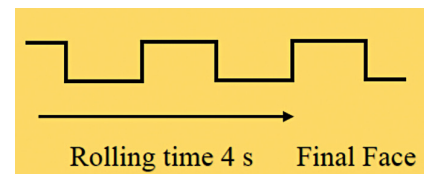


Fig. 3: Timing diagram of the e-dice throw

Components	Quantity	Description
ESP32 board	1	For programming
74LS138	1	To decode the random 3-bit number
74LS260	1	To set the correct combinational logic
CD4075	1	To set the correct combinational logic
74LS04	1	To set the correct combinational logic
5mm LEDs	7	To display a valid face of the dice
330Ω resistors	7	Series resistors for LEDs to limit current
Jumpers	As required	Male-to-female jumper wires for connection

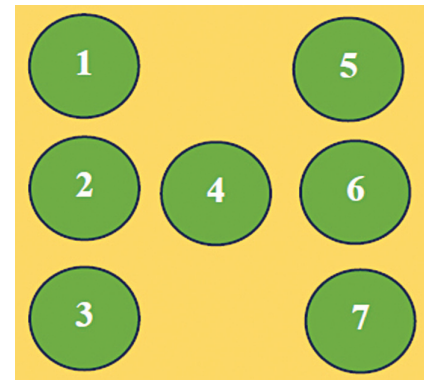


Fig. 4: Arrangement of LEDs for the e-dice